GRAD-RL: Graph-Augmented Hybrid AI Architecture for SCADA Resilience

Autonomous Cyber-Physical Defense: A Graph-Augmented Hybrid AI Architecture for SCADA Resilience (Thesis Companion)

ICS / SCADA Telemetry



- Process variables
- Network data
- Asset states

GRAD-RL Pipeline

K1 Anomaly Detection (LSTM Autoencoder)



- Detects deviations from learned normal behavior
- Reconstruction error $e > \tau$

K2 Technique Classification (XGBoost)



- Maps to MITRE ATT&CK for ICS techniques
- Outputs class probabilities

K3 Graph Risk Analysis (Differentiable Graph Tensor)



- Uses graph position (betweenness)
- Weights with CVSS v4.0 severity
- $R(v,t)$ tensor, normalized to 0–10

K4 Response Agent (PPO)



- Selects response action
- MDP with state from K1–K3
- Constrained by safety gate

Dual Safety Gate

IEC 61511 + IEC 62443

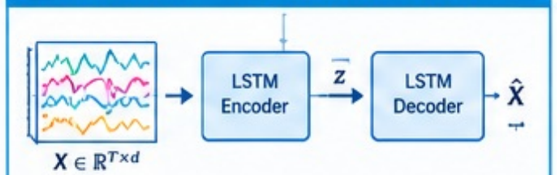
- $R \geq 9.0$ (CVSS Critical)
- Process safety + security
- Approve / Replace / Operator
- AND or OR (to be defined)

SOC & Containment



- Dragos (OT)
- Claroty (OT)
- Nozomi (OT)
- IBM QRadar (SIEM)
- Splunk (SIEM)
- CrowdStrike Falcon (Endpoint)

K1 LSTM Autoencoder (Perception Layer)



Mechanism

- Learns normal temporal & cross-variable structure
- Reconstruction error $e = \|X - \hat{X}\|$
- Anomaly if $e > \tau$

GRAD-RL specifics

- $\tau = 0.001607$ (locked)
- Validation loss = 0.000277
- Blind to attacks that leave process variables unchanged (low control-layer recall on MaCySTe)

K2 XGBoost Technique Classifier



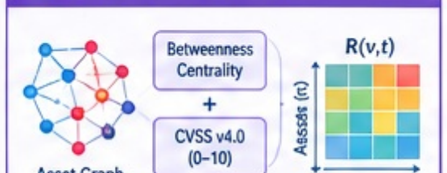
Mechanism

- Gradient-boosted decision trees
- Softmax (multiclass), argmax
- Regularization, depth limits, shrinkage
- Unseen techniques → nearest known class

GRAD-RL specifics

- Classes: MITRE ATT&CK for ICS + *Normal*
- OT lab: 519 events (Normal 429, T0855 30, T0813 30, T0856 30)
- MaCySTe: 5 attack classes, 3,249 / 93,502
- Per-class F1 (MaCySTe): T0856 0.6424, T0814 0.6607

K3 Graph Risk Tensor



Mechanism

- Betweenness = propagation / process impact
- CVSS v4.0 = asset severity (0–10)
- $R(v,t)$ tensor over asset and time
- Normalized to CVSS scale (gate threshold 9.0)

GRAD-RL specifics

- Complexity: $O(V \cdot E)$ (Brandes-style)
- Differentiable parts and exact $R(v,t)$ form to be specified in thesis (Sec. 11)

K4 PPO Response Agent



Mechanism

- On-policy actor-critic
- Clipped surrogate objective
- MDP with state from K1–K3

GRAD-RL specifics

- State vector, action set, reward terms, discount factor, PPO hyperparameters
- GAM: surjective mapping (see Section 11)

Dual Safety Gate



Mechanism

- Rule-level control (not learnable)
- Considers security criticality + process safety consequence

GRAD-RL specifics

- Threshold: 9.0 (corrected from 7.0)
- MaCySTe live: 57/63 → SL4, precision 1.0
- Branch logic and AND/OR combination to be specified

Design Rationale

Topological blindness

- Per-asset detectors treat all assets equally. K3 uses graph position to weight risk.

Temporal asymmetry

- Attackers act at machine speed.
- K4 enables autonomous response constrained by safety gate.

T0856 Sensor Spoof (MITRE T0856)

- Tank level constant (50.0) while pressure escalates
- Each value is individually plausible → range checks / signature IDS do not fire
- Cross-variable physical relationship violated (analogous to Stuxnet replay)

Experimental Evidence

Setting	Result
Locked canonical metrics	F1 0.95 / 0.89, latency 45 ms, containment 91.3%, τ 0.001607, $O(V \cdot E)$
GRFICSv3 offline	99.70% accuracy
SWaT	99.16%
WADI	74.17%
HAI	F1 0.605
GRFICSv3 live	10,352 rows / 111.8 min, 18 scenarios, FPR 0%, latency 9.7 ± 7.2 s, 10/10 detected
MaCySTe offline	98.28% acc, FPR 0.88%, T0856 0.6424, T0814 0.6607
MaCySTe live	57/63 → SL4, precision 1.0

Known Limitations

- MaCySTe NATS KV attacks (T0831, T0836, T0827) do not change sensor values (physics engine ignores writes) → ~5% recall on control-layer attacks.
- Lower performance on WADI and HAI vs. GRFICSv3 and SWaT.
- Rare attack classes in OT lab dataset (30 events each) limit per-class F1 statistical weight.
- K2 cannot recognize techniques absent from training data

Parameters to Specify

Item	Stage
Window length T, feature set d, layer sizes, latent dimension, τ rule	K1
Feature set, K1-only?, class balancing, tree hyperparameters	K2
$R(v,t)$ equation, CVSS groups, graph construction, differentiable ops	K3
State vector, action set, reward terms, discount factor, PPO hyperparameters, GAM	K4
Gate inputs, per-standard conditions, branch logic, operator approval, AND/OR	Gate
SWaT/WADI metric definitions, dataset for F1 values	Evaluation

GRAD-RL: Technical Architecture Description

Graph-Augmented Hybrid AI Architecture for SCADA Resilience. Companion document to the thesis "Autonomous Cyber-Physical Defense: A Graph-Augmented Hybrid AI Architecture for SCADA Resilience".

1. System Overview

GRAD-RL is a four-stage pipeline for autonomous defense of ICS/SCADA environments. Stage K1 detects deviations from learned normal process behavior, K2 attributes the deviation to a MITRE ATT&CK for ICS technique, K3 quantifies the topological and vulnerability-weighted risk of the affected assets, and K4 selects a response action. A dual safety gate based on IEC 61511 and IEC 62443 constrains the action before it reaches SOC and containment tooling.

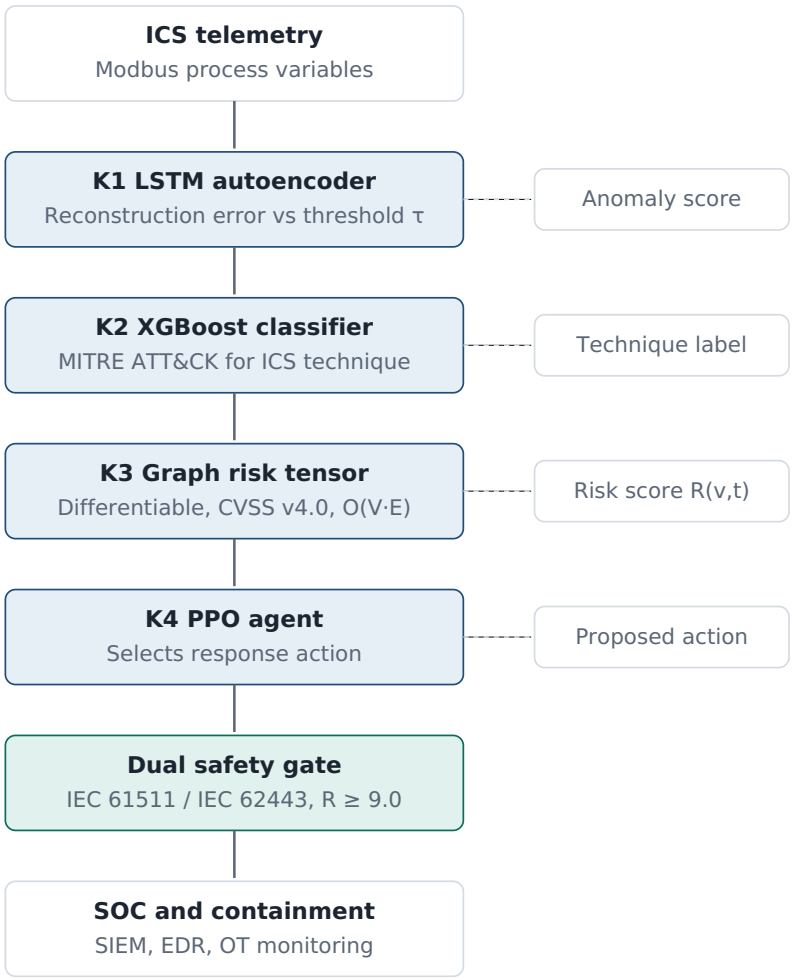


Figure 1. End-to-end GRAD-RL data flow. Stage ordering follows the K1 to K4 numbering and the gate placement before SOC tooling; conditional gating between stages is not drawn because it is not specified in the source notes.

2. Design Rationale

Two weaknesses motivate the architecture. The first is topological blindness: per-asset anomaly detectors treat all assets as equally consequential, although a compromise of a bridging controller propagates differently from a compromise of a peripheral sensor. K3 addresses this by weighting risk with graph position. The second is temporal asymmetry: attackers act at machine speed while manual triage does not, which motivates the autonomous K4 layer constrained by a safety gate.

The central detection argument rests on the T0856 sensor spoof. The attacker holds the tank level register constant at 50.0 while pressure escalates. Each value is individually plausible, so range checks and signature-based IDS do not fire. The cross-variable physical relationship is violated, and this is what K1 learns to detect. The scenario is analogous to the Stuxnet replay of normal sensor values.

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3. K1: LSTM Autoencoder (Perception Layer)

Mechanism

The input is a sliding window of multivariate process data, X of shape T×d, with T time steps and d process variables. An LSTM encoder compresses the window into a latent vector z, and an LSTM decoder reconstructs $\hat{X}$ from z. The model is trained on normal data only, so it learns normal temporal and cross-variable structure. The reconstruction error $e = \|X - \hat{X}\|$ increases when this structure is broken. The decision rule is anomaly if $e > \tau$.

GRAD-RL specifics

The locked threshold is $\tau = 0.001607$. The offline MaCySTe validation loss of the autoencoder is 0.000277. K1 is structurally blind to attacks that leave process variables unchanged, which explains the weak control-layer recall on MaCySTe (Section 10).

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4. K2: XGBoost Technique Classifier

Mechanism

XGBoost builds an additive ensemble of decision trees, each fitted to the gradient and second derivative of the loss with respect to the current ensemble prediction. In the multiclass setting a softmax objective yields one tree sequence per class and a class probability vector, with the predicted class taken as the argmax. Regularization, depth limits, and shrinkage control overfitting. The classifier can only emit classes seen in training, so unseen techniques are mapped to the nearest known class.

GRAD-RL specifics

Classes are MITRE ATT&CK for ICS technique identifiers plus a normal class. The OT lab dataset contains 519 events (Normal 429, T0855 30, T0813 30, T0856 30). The MaCySTe dataset contains five attack classes, with 3,249 attack rows among

93,502. Per-class F1 on MaCySTe offline is 0.6424 for T0856 and 0.6607 for T0814, well below overall accuracy because normal traffic dominates.

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5. K3: Differentiable Graph Risk Tensor

Mechanism

K3 combines asset position in a directed graph with vulnerability severity. Betweenness centrality measures how many shortest paths traverse a node, which approximates propagation and process impact if the node is compromised. CVSS v4.0 supplies asset-specific severity on a 0 to 10 scale. The resulting risk R(v,t) is indexed by asset v and time t, so risk is maintained as a tensor over both dimensions and normalized to the CVSS scale because the safety gate threshold is 9.0.

GRAD-RL specifics

The locked complexity is O(V·E), consistent with Brandes-style betweenness computation. Which operations in the formulation are differentiable, and the exact form of R(v,t), are items to be stated explicitly in the thesis (Section 11), since betweenness itself is combinatorial and does not propagate gradients.

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6. K4: PPO Response Agent

Mechanism

Proximal Policy Optimization is an on-policy actor-critic method that maximizes a clipped surrogate objective, which limits the size of each policy update relative to the previous policy. The response problem is formulated as a Markov decision process whose state is derived from K1 to K3 outputs.

GRAD-RL specifics

The DRL paper contains an MDP formulation and a GenericAssetMapper (GAM) surjective mapping. The action space is mapped to SOC tooling (Section 8). The explicit state vector, action set, reward terms, discount factor, and PPO hyperparameters are not specified in the source notes and are listed in Section 11 rather than assumed here.

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7. Dual Safety Gate

Mechanism

The gate sits between the learned policy and execution. It is a rule-level control independent of the policy, so the agent cannot learn to bypass it. IEC 61511 addresses functional safety in process industries and IEC 62443 addresses industrial cybersecurity, and the dual construction requires consideration of both the security criticality and the process safety consequence of an action.

GRAD-RL specifics

The activation threshold is R ≥ 9.0, aligned with the CVSS Critical band. The

threshold was corrected from 7.0 to 9.0 during journal revision, and the associated design requirement DR4 was narrowed to critical-only. On MaCySTe live evaluation, 57 of 63 detections escalated to IEC 62443 Security Level 4 with precision 1.0. The branch logic (approve, replace, or route to operator) and whether the two standards are combined by AND or OR are to be specified.

8. SOC Integration

The K4 action space maps to commercial and open tooling: Dragos, Claroty, and Nozomi on the OT monitoring side, IBM QRadar and Splunk on the SIEM side, and CrowdStrike Falcon on the endpoint side. The gate output is the interface to these systems.

9. Experimental Evidence

Setting	Reported result
Locked canonical metrics	F1 = 0.95 / 0.89, latency 45 ms, containment 91.3%, τ = 0.001607, O(V·E)
GRFICSv3 offline	99.70% accuracy
SWaT	99.16%
WADI	74.17%
HAI	F1 = 0.605
GRFICSv3 live	10,352 telemetry rows over 111.8 min, 18 attack scenarios, decision-level FPR 0%, mean detection latency 9.7 ± 7.2 s, 10 of 10 process-impacting attacks detected
MaCySTe offline (macyste_dataset_v2.csv, 91,766 rows)	Accuracy 98.28%, FPR 0.88%, T0856 F1 0.6424, T0814 F1 0.6607
MaCySTe live	57 of 63 detections escalated to IEC 62443 SL4, precision 1.0

10. Known Limitations

On MaCySTe, attacks written through NATS KV (T0831, T0836, T0827) do not change any sensor value because the BridgeCommand physics engine ignores those writes, which yields roughly 5% recall on control-layer attacks. This is a design boundary of a process-variable detector, not a tuning issue. Performance on WADI and HAI is substantially lower than on GRFICSv3 and SWaT. Rare attack classes in the OT lab dataset contain 30 events each, which limits the statistical weight of per-class F1. K2 cannot recognize techniques absent from training data.

11. Parameters to Specify

Item	Stage
Window length T, feature set d, layer sizes, latent dimension, derivation rule for τ	K1
Feature set, whether K2 runs only on K1-flagged windows, class balancing method, tree hyperparameters	K2
Exact R(v,t) equation, CVSS v4.0 metric groups used, graph construction, which operations are differentiable	K3
State vector, action set, reward terms, discount factor, PPO hyperparameters, GAM definition	K4
Gate inputs, per-standard conditions, branch behavior for R ≥ 9.0 and R < 9.0, operator approval step, AND or OR combination	Gate
Metric definitions for the SWaT and WADI values, and which dataset each locked F1 value (0.95, 0.89) belongs to	Evaluation